

S175 simulator
full factorial dataset: 126,153,720 cases
(H_s refined to 0.25 m, χ to 5°)

Stratified split over distinct
input combinations (80/10/10)
+ structured holdout regimes

Model-family screening on a
stratified 5% subsample:
RF / XGBoost / MLP → MLP selected

Stage 1 — feasibility classification

C_1 : complete infeasibility
(all outputs = -1)

C_2 : fuel-only infeasibility
(only fuel = -1)

thresholds τ_1, τ_2 : most selective with
validation recall ≥ 0.99

Stage 2 — multi-output MLP regressor (512-256-128), masked asymmetric loss
trained on fully valid + fuel-only infeasible cases; fuel term masked where undefined

Sweep 1 (subsample): uniform
 $\alpha \in \{1.0, 1.5, 2.0, 3.0, 5.0\}$
→ $\alpha_{\text{speed}} = 1.5$

Sweep 2 (subsample): others
 $\alpha \in \{1.5, 0.8, 0.67, 0.5, 0.33\}$
→ $\alpha_{\text{others}} = 0.33$ (10/10 safe side)

Final regressor: masked loss, per-output weights ($\alpha_{\text{speed}} = 1.5, \alpha_{\text{others}} = 0.33$)
every output biased toward its safe error side

Assembled two-stage surrogate
 $C_1 \geq \tau_1 \rightarrow$ **all -1**; **else regressor**; $C_2 \geq \tau_2 \rightarrow$ **fuel = -1**

Evaluation: held-out test set · structured holdouts
· fresh off-grid simulator runs · interpolation baseline